

Mini Project Assignment

Design and Development of a Complete Database System for a Forensic Medical Department

For a Forensic Medicine Department complete database system project, a group of **4 students** is ideal.

1. Background

Currently, many forensic medical departments maintain records manually using paper-based files. This creates difficulties in:

- Searching patient/case records
- Maintaining confidentiality
- Generating court reports
- Tracking evidence and samples
- Managing medico-legal documents
- Producing statistical reports

Students are required to analyze the existing manual process and develop a complete computerized database system for a forensic medical department.

2. Project Objectives

The objectives of this project are to:

- Understand real-world database design
- Practice requirement gathering and system analysis
- Design conceptual, logical, and physical database models
- Implement a relational database
- Develop a user-friendly application interface
- Generate medico-legal reports
- Apply normalization and database security concepts

3. Expected System Modules

Each group should develop a complete database system including the following modules.

Core Modules

1. **Patient Management**
 - Register patients
 - Maintain personal details
 - Track patient history
2. **Case Management**
 - Create medico-legal cases
 - Assign case numbers
 - Record incident details
3. **Postmortem Management**
 - Record autopsy details
 - Store cause of death
 - Maintain examination reports
4. **Evidence Management**
 - Track forensic samples
 - Maintain chain of custody
 - Store laboratory evidence information
5. **Court Report Management**
 - Generate medico-legal reports
 - Produce court submission documents
 - Track report status
6. **Staff Management**
 - Doctors
 - Judicial Medical Officers
 - Laboratory staff
 - Clerical officers
7. **User Authentication and Security**
 - Login system
 - User roles
 - Access control
8. **Report Generation**
 - Daily case reports
 - Monthly statistics
 - Pending case reports

4. Minimum Functional Requirements

The system should support:

- Add, update, delete, and search records
- SQL queries and reports
- Data validation
- Multi-table relationships
- Backup and recovery concepts

- Role-based access control

5. Additional Features

- Barcode/QR code integration
 - Audit logging
 - Dashboard and analytics
 - Mobile responsiveness
 - Email notification system
 - Digital signature support
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A recommended structure is:

For your assignment, **4 students per group** would work very well because the project includes:

- Requirement analysis, Documentation, Presentation
 - Database design, Testing & Quality Assurance Member
 - SQL implementation
 - UI / Front-End Developer
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Suggested Group Member Roles and Duties

1.1 Project Coordinator / Team Leader

Responsibilities

- Coordinate team activities
- Manage deadlines
- Communicate with lecturer
- Organize meetings
- Combine final submissions

Deliverables

- Project schedule
 - Progress tracking
 - Final compiled report
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1.2 System Analyst / Requirement Analyst

Responsibilities

- Study current manual system
- Collect requirements
- Identify stakeholders
- Prepare Software Requirement Specification (SRS)

Deliverables

- Requirement analysis document
- Functional requirements
- Non-functional requirements
- Workflow descriptions

1.3 Documentation

Responsibilities

- Prepare final report
- Organize screenshots and diagrams
- Prepare presentation slides
- Coordinate final demonstration

Deliverables

- Final documentation
- Presentation slides
- Demo preparation

2.1 Database Designer

Responsibilities

- Identify entities and relationships
- Create ER diagrams
- Design relational schema
- Normalize tables

Deliverables

- ER Diagram
- Relational schema
- Data dictionary
- Normalization steps

2.2 Testing & Quality Assurance Member

Responsibilities:

- Test SQL queries
- Verify relationships
- Check constraints
- Validate reports

3.1 Database Developer / SQL Programmer

Responsibilities

- Create database tables
- Implement constraints
- Insert sample data
- Write SQL queries, views, procedures, triggers

Deliverables

- SQL scripts
- Database implementation
- Query outputs

3.2 Database Requirements

Students must include:

Conceptual Design

- Entity Relationship Diagram (ERD)

- EER diagram if applicable

Logical Design

- Relational schema
- Primary and foreign keys
- Constraints

Physical Design

- Table creation scripts
- Indexes
- Views

Database Implementation

Use one of the following:

- MySQL
- PostgreSQL
- MariaDB
- SQLite

3.3 Suggested Entities

Students may identify additional entities during requirement analysis. Following are main entities.

Entity	Example Attributes
Patient	PatientID, Name, Age, Gender, Address
Case	CaseID, CaseType, IncidentDate
Doctor	DoctorID, Name, Specialization
Postmortem	PostmortemID, Findings, CauseOfDeath
Evidence	EvidenceID, EvidenceType, StorageLocation
CourtReport	ReportID, SubmissionDate, Status
LaboratoryTest	TestID, TestType, Result
Staff	StaffID, Role, ContactNo
User	Username, Password, UserRole

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4. Web/Application Developer (UI / Front-End Developer)

Responsibilities

The student should develop:

- Login page
- Data entry forms
- Data entry screens
- Search interfaces
- Report pages/ interfaces
- Navigation menus
- Database connectivity

Typical Modules to Develop

Module	Purpose
Login System	User authentication
Patient Registration	Add patient details
Case Management	Add/manage forensic cases
Examination Form	Record examination findings
Report Generation	Generate medico-legal reports
Search System	Retrieve case information
Dashboard	Display statistics

Frontend

- HTML/CSS/JavaScript
- PHP / Python Flask / Django / Java

Optional

- Bootstrap
- Chart libraries
- REST APIs

Technologies Students Can Use

- PHP + MySQL
 - Python Flask/Django
 - Java Spring Boot
 - Node.js
 - HTML/CSS/JavaScript
 - Bootstrap
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- Flask + MySQL is modern and educational

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Optional

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- Chart libraries
- REST APIs

5.Additional Features

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Suggested Task Allocation Table

Student Role	Main Tasks/ Responsibilities
Team Leader Documentation	Coordination & submission ,Report & testing
Requirement Analyst ,Testing	Requirement gathering & SRS
Database Designer	ER diagrams, schema & normalization
Web Interface Developer	Forms & web system

Good Practice for Assessment

Individual Contribution Report

Each student should mention:

- Tasks completed
- Contribution percentage
- Challenges faced

This helps evaluate individual participation fairly.

Example Instruction to Students

Students should work in groups of 4 members.

Each member must be assigned a clearly defined role and responsibilities.

The final report must include an individual contribution section describing the work completed by each member.

Recommended Project Timeline

Week	Activity
Week 1	Requirement gathering
Week 2	ER design & Web interface design
Week 3	Relational design & normalization , Interface development, test with use access authentications
Week 4	SQL implementation + Implementation and link software interface
Week 5	Queries & reports

Week	Activity
Week 6	Documentation & presentation

Assessment Scheme

Component	Marks
Requirement Analysis	10
ER Diagram & Design Normalization & Schema	15
Database Implementation	30
Frontend/Backend Development	25
Queries & Reports	10
Presentation & Demo	10
Total	100

Additional Requirements

Students should:

- Use proper normalization techniques
- Apply constraints and referential integrity
- Design meaningful reports
- Consider confidentiality of forensic records
- Include sample data for testing
- Demonstrate backup and recovery procedures

Final Demonstration

During the final evaluation, each group must:

- Demonstrate all system functions
- Explain database design decisions
- Execute SQL queries
- Show generated reports
- Answer viva questions

